

HOW HEADPHONES USE MATH TO KILL NOISE.



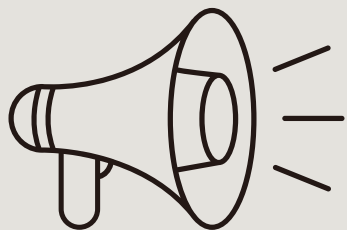
The math and physics behind
noise Cancellation



NEXT >

The Problem.

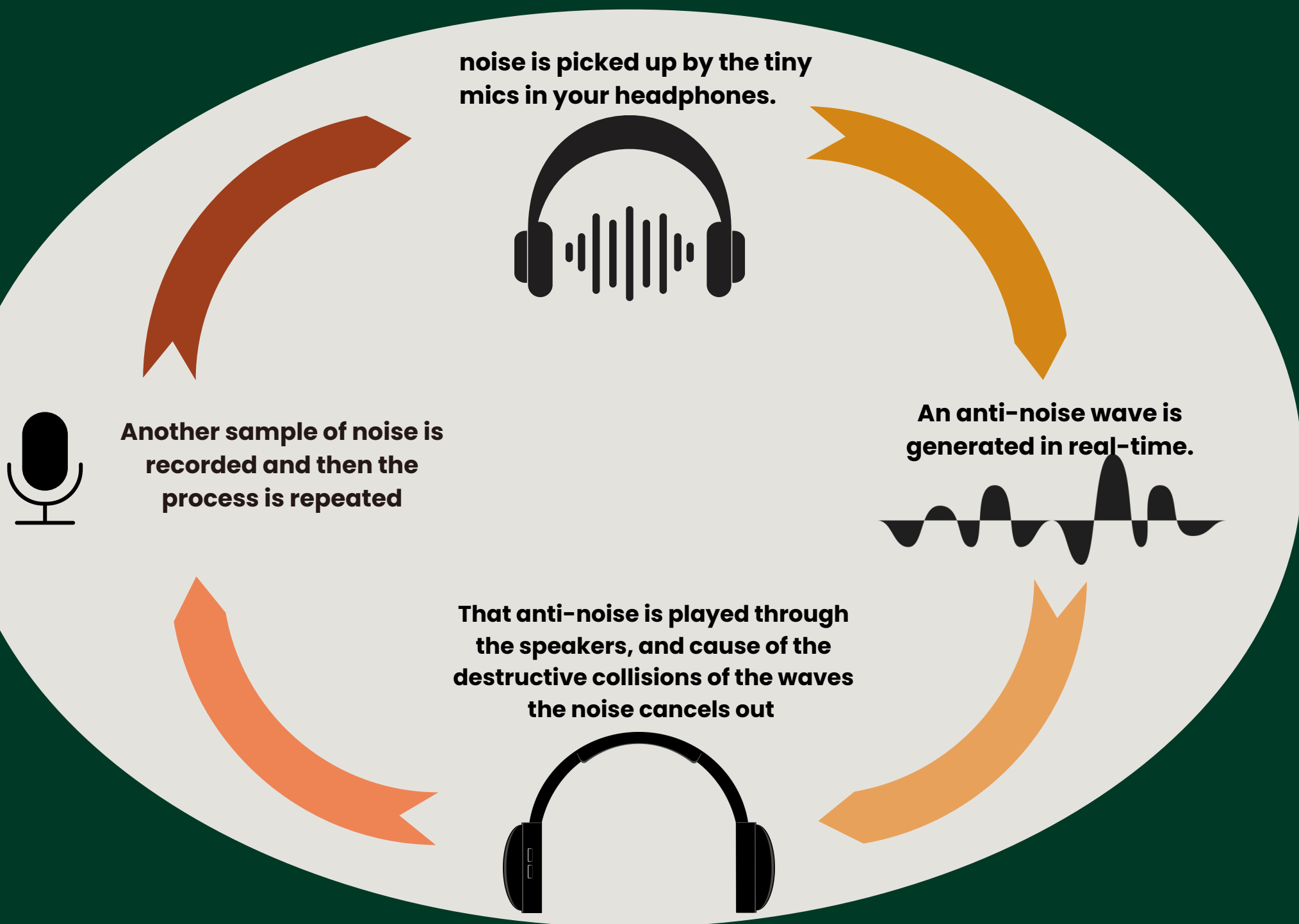
There's noise coming from all directions, from planes overhead, traffic outside, a humming AC unit, dogs barking, and even a loud group of friends chatting in the library.



NEXT >

Active Noise Cancellation.

A technology that reduces unwanted sound by generating an opposite sound wave (anti-noise) that cancels out the original noise through destructive interference, all this must be done in real time.



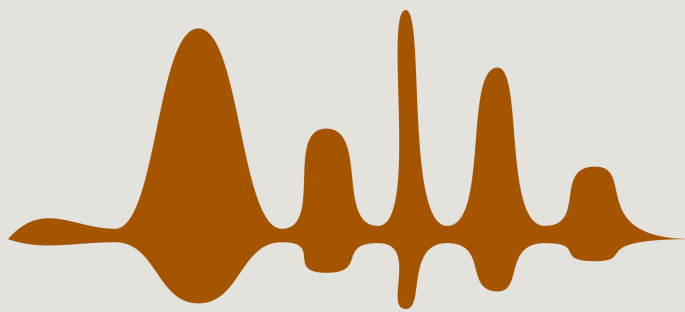
How the anti-noise is made.

DSP (digitalized Sound Data) is used to invert the sound waves that are received from the mic.

An inversion of a sound wave is exactly 180 degrees opposite to the original sound wave.

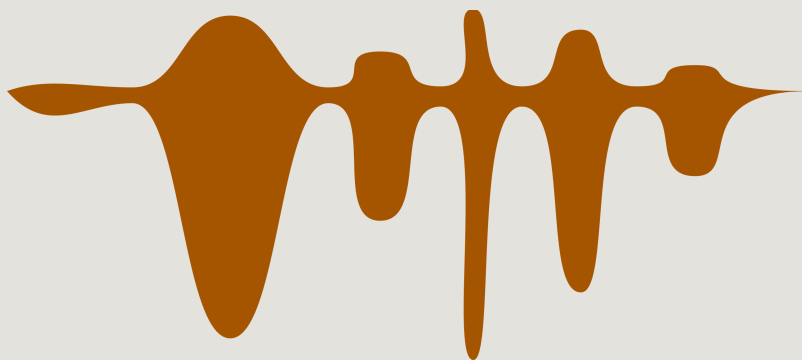
Noise

$$f(\text{Original Wave}) = A \times \sin(w)$$



anti-noise

$$f(\text{anti Wave}) = -A \times \sin(w)$$



$$f(\text{new}) = A \times \sin(w) + (-A \times \sin(w)) = 0$$

Silence



This Destructive collision causes the noise to cancel out

The Significance of Active Noise Cancellation.

Active Noise Cancellation isn't just a comfort feature, it's a powerful example of how math, engineering, and real-time computing come together to solve everyday problem.

From:

- Improving focus in noisy environments
- Reducing fatigue during travel
- Enhancing call clarity
- Protecting long-term hearing

...it's a quiet revolution happening in our ears, driven by precision wave manipulation and clever signal processing



IN CLOSING

The next time you switch on ANC and the world fades into silence, remember, It's wave physics, fast DSP, and clever engineering working in harmony.

🗣️ Curious: What's another everyday tech feature that quietly amazes you?

